CampusIQ - Day 2 Notes

Name: Gufran Date: 19 October 2026 Role: AI (/ai folder)

Topic: Virtual environment, reading files, chunking, keyword search

0. Why chunking? (the concept)

- College PDFs are 50-100 pages long. An AI model can read only a limited amount of text at once (context window).
- Giving the whole PDF is slow, costly and makes the right line hard to find.
- So we split the text into small pieces called chunks. For a question we give the model only the 3-4 most relevant chunks.
- This is the first step of RAG (Retrieval-Augmented Generation): split, then search, then answer.
- Each chunk keeps its topic (like "Attendance Rules"), so we always know where a rule came from.

1. Virtual environment (venv)

- A **venv** is a separate box of Python libraries, only for this project. It stops different projects from clashing.

```
cd ai
python -m venv .venv              # create it (once)
.venv\Scripts\activate            # turn it on (every new terminal)
```

- It works when the prompt starts with (.venv).
- If PowerShell blocks it ("running scripts is disabled"), run once: `Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned`
- The `.venv` folder is never pushed to GitHub (it is in .gitignore).

2. Installing a library with pip

```
pip install pymupdf
python -c "import pymupdf; print('ready')"
```

- pip = Python's app store. PyMuPDF reads text out of PDFs. We will use it on the real college PDFs.
- Write `import pymupdf` (the old name `fitz` still works but is deprecated).
- The "new pip release" notice is only a suggestion. Ignore it.

3. Reading a text file

```
with open("data/sample_rules.txt", "r", encoding="utf-8") as f:
    text = f.read()
lines = text.split("\n")
print(len(text), len(lines))      # 795 characters, 17 lines
```

- `open` opens the file. `"r"` = read only. `with` closes the file automatically.
- `encoding="utf-8"` makes sure all characters are read correctly.
- `split("\n")` cuts text at every new line and returns a list of lines.
- The path `data/...` is relative to the folder where I run Python, so the terminal must be in `ai`.

4. My first chunker

```
chunks = []
current_topic = ""
for line in lines:
    line = line.strip()
    if line == "":
        continue                          # skip blank lines
    if line.endswith("Rules"):
        current_topic = line              # a heading: remember it
    else:
        chunks.append({
            "id": len(chunks) + 1,
            "topic": current_topic,
            "text": line,
        })
```

- Every rule line becomes one chunk: a dictionary with `id`, `topic` and `text`.
- Result: 9 chunks (3 Attendance, 3 Examination, 3 Hostel).
- `strip()` removes extra spaces. `continue` jumps to the next line.

5. Saving to JSON

```
import json
with open("data/chunks.json", "w", encoding="utf-8") as f:
    json.dump(chunks, f, indent=2)
# later: chunks = json.load(f)
```

- JSON = standard text format for lists and dictionaries. `"w"` = write (replaces the old file).
- The same shape `[{id, topic, text}]` is what embeddings and the vector database will use later.

6. Keyword search and why it fails

```
def words(text):
    cleaned = text.lower().replace("?", "").replace(".", "")
    return {w for w in cleaned.split() if w not in STOP_WORDS}

score = len(question_words & chunk_words)   # common words
scored.sort(key=lambda pair: pair[0], reverse=True)
```

- A **set** keeps unique words. `&` gives the words common to both sets. The count is the score.
- **Stop words** ("the", "is", "of") are ignored. `sort(..., reverse=True)` puts the highest score first.

Question	Result
How much attendance is needed for the exam?	Found the attendance rule.
When can visitors come to the hostel?	Right chunk, but 3-way tie. It came 3rd by luck.
Minimum presence required to sit in finals?	All scores 0. No common words, so no understanding.

Big lesson: Keyword search matches words, not meaning. "presence" = "attendance" and "finals" = "end semester exam" for a human, but not for this code. The fix is **embeddings** (Day 3): numbers that place similar meanings close together.

7. My mistakes today

What happened	Reason and fix
`can't open file day2_read.py`	Terminal was in `campusiq`, the file is in `ai`. Run `cd ai` first.
`git add .` from inside `ai`	It adds only that folder. Run git commands from the repo root.
`push rejected (fetch first)`	My friend pushed first. Run `git pull`, then `git push`.
Vim opened after `git pull`	It asks for a merge message. Press Esc, type `:wq`, press Enter.

8. Self-check (answers)

- Why chunk a document? The model reads limited text at once, and small relevant pieces give better, cheaper answers.
- What is a venv? A separate library box for one project, active when `(.venv)` shows.
- Why does Keyword search fail? It needs the same words and cannot match meaning.
- Why rejected push? The remote had commits I did not have. Pull first, then push.

9. Quick revision sheet

```
ENV     cd ai ; .venv\Scripts\activate ; pip install NAME
FILES   with open(path, "r", encoding="utf-8") as f: text = f.read()
LISTS   text.split("\n") -> list ; list.append(x) ; for x in list:
JSON    json.dump(obj, f, indent=2)  /  obj = json.load(f)
SETS    a & b = common items ; len(a & b) = how many
GIT     repo root: pull, add ., commit -m "", push
```

10. Files created today

- `ai/data/sample_rules.txt` - practice rules (3 topics, 9 rules)
- `ai/day2_read.py`, `day2_chunks.py`, `day2_search.py`
- `ai/data/chunks.json` - 9 chunks saved

Day 2 complete. Next (Day 3): embeddings. Turn text into vectors so "presence" matches "attendance". We will also install the sentence-transformers library.